CHESS MOVES AND DISTANCE ON ONE PAGE

Rook, king and crow

Three ways to measure a journey across the board, and the shortest paths that each one allows.

THE RULE

One displacement, three distances

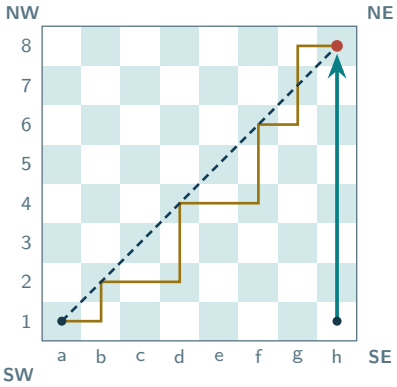
Give each square coordinates, files a to h as $x = 1$ to 8 and ranks as $y = 1$ to 8, and write a journey as the displacement $(\Delta x, \Delta y)$. The three measures are the cases $p = 1, 2, \infty$ of $d_p = \big(|\Delta x|^p + |\Delta y|^p\big)^{1/p}$:

rook, one square at a time	$d_1 = \lvert\Delta x\rvert + \lvert\Delta y\rvert$
crow, in a straight line	$d_2 = \sqrt{\Delta x^2 + \Delta y^2}$
king, counting moves	$d_\infty = \max\big(\lvert\Delta x\rvert, \lvert\Delta y\rvert\big)$

The rook's d_1 is the Manhattan or taxicab distance and the king's d_∞ is the Chebyshev distance, and for every journey

$$d_\infty \;\le\; d_2 \;\le\; d_1 \;\le\; 2\,d_\infty,$$

with all three equal exactly when the journey runs along a single rank or file.



SE to NE along the h-file, and SW to NE with a rook staircase and the crow's line.

TWO JOURNEYS TO THE NE CORNER

Along an edge the measures agree, across the diagonal they split

From h1 in the SE corner to h8 in the NE corner the displacement is $(0, 7)$, so all three distances are 7; from a1 in the SW corner the displacement is $(7, 7)$, and the three measures separate as far as the inequality allows. The geodesic counts, the number of different shortest paths, behave in the opposite way to the distances.

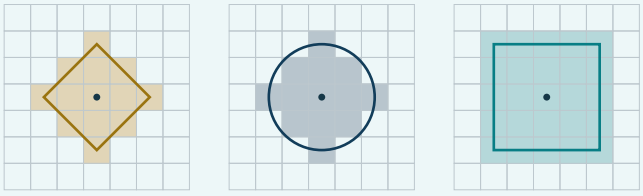
journey	d_1	d_2	d_∞	shortest rook-step paths	shortest king paths
h1 to h8, SE to NE	7	7	7	1	127
a1 to h8, SW to NE	14	$7\sqrt{2} \approx 9.90$	7	$\binom{14}{7} = 3432$	1

A real rook covers either journey in at most two moves, since it slides any number of squares; its path *length* is d_1, while its move *count* is 0, 1 or 2.

CIRCLES

Every square within distance 2

A circle is the set of points at a fixed distance from a centre, so each measure draws its own circle. The rook's is a diamond, the king's is a square and only the crow's is round; the king's square of radius 2 holds 25 squares, the rook's diamond 13, and these are exactly the squares a king reaches in two moves and a rook reaches in two single steps. At radius 2 the crow's squares happen to coincide with the rook's diamond, and the two part company from radius 3, where d6 to f8 lies within $2\sqrt{2}$ by air but 4 by rook.



Counting the shortest paths

A geodesic is a path whose length equals the distance. The crow's geodesic is the unique straight segment, while a rook-step geodesic from a1 to h8 is any order of 7 steps right and 7 steps up, giving $\binom{14}{7}$. A king's geodesic up the h-file makes 7 moves upward, each drifting one file left, right or not at all, without leaving the board and returning to the h-file, which is counted by the Motzkin number $M_7 = 127$.

Geodesics on a curved surface

On a flat board the crow's geodesic is a straight line, but on the Earth the shortest route between two places is an arc of a great circle, with length $R\,\Delta\sigma$ for the central angle $\Delta\sigma$; flying between two cities on the same latitude therefore bends towards the pole, which a flat map draws as a curve.

Conventions. Distances are measured between square centres in units of one square, with White's side at the south, so a1 is SW, h1 is SE and h8 is NE. A rook-step means a rook moving one square at a time along a rank or file, and d_∞ is the limit of d_p as $p \to \infty$.

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